

Introduction to Colab

Google has provided a free cloud service called COLAB that is based on Jupyter Notebooks that supports free GPU (for a number of compute units). It is a great tool for improving your coding skills, and it allows you to develop deep learning applications using popular libraries such as **PyTorch**, **TensorFlow**, **Keras**, and **OpenCV**.

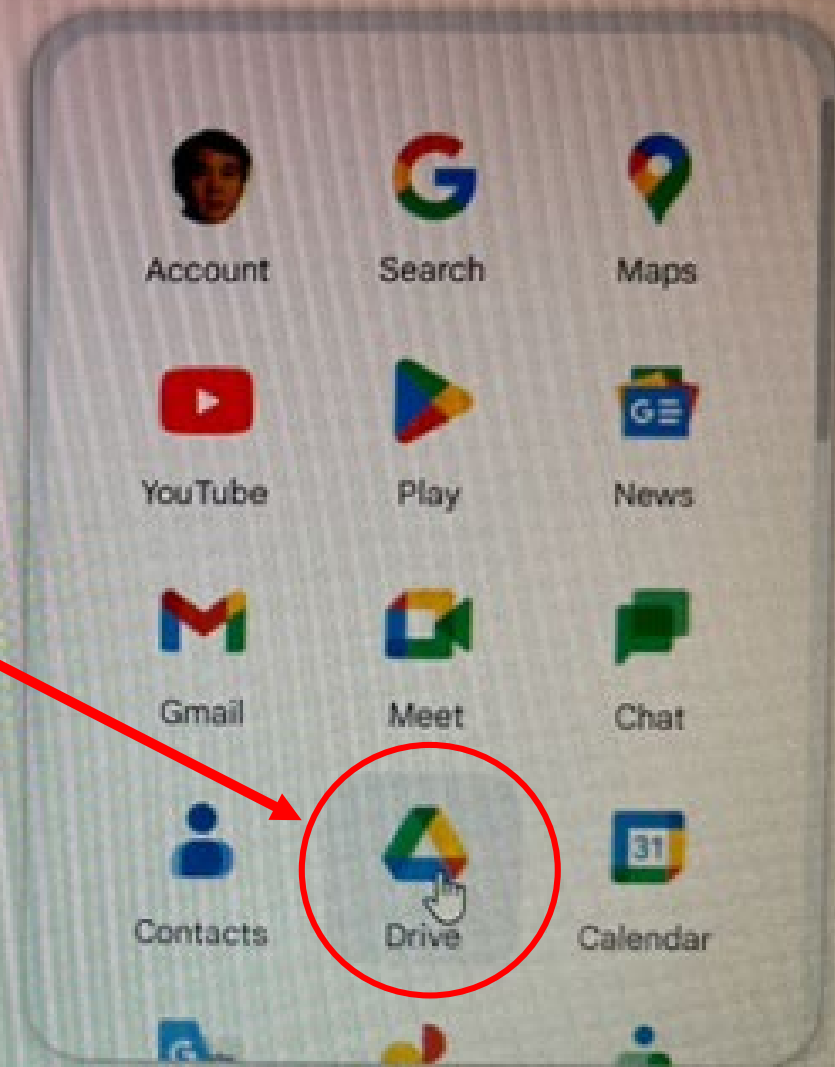
To start, go to your Google Drive, and learn to create a python file for Colab to run.

<https://www.datasciencecentral.com/profiles/blogs/all-about-using-jupyter-notebooks-and-google-colab>

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Getting started



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Folder upload



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These apps can be used directly from Google Drive to create or edit files.

Scrible, Inc.

Search and bring your research into Google Docs while automagically managing your...

★ 2.4 • 10,000,000+



Scrible Video Edit...

[scrible-ext-dev](#)

Browser-based video editor with the simplicity you expect from Scrible. Integrates...

★ 3.8 • 10,000,000+

draw.io

draw.io is the most tightly Google Drive integrated diagramming application...

★ 4.6 • 10,000,000+



Adobe Acrobat - PDF a...

[Adobe](#)

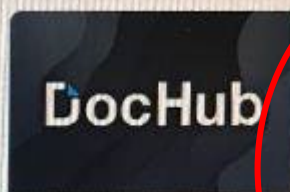
Get the world's best PDF and e-signature tools embedded inside Google Drive

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★ 4.5 • 9,800,788

www.smartsheet.com

Smartsheet

★ 3.6 • 10,000,000+



Colaboratory

[colab-team](#)

Colab is a Jupyter notebook environment that runs in the browser using Google Cloud.

★ 4.7 • 9,300,389



Google Apps Script

Google Apps Script

[Google](#)

Google Apps Script is a JavaScript platform that provides easy ways to integrat...

★ 4.4 • 9,182,163



MindMup 2 For Google

[MindMup](#)

Collaborative Mind mapping, tightly integrated with Google Apps

★ 4.3 • 8,552,587

Lucidpress

Lucidpress K12

[Lucid Software](#)

Whether you're a teacher or student, with Lucidpress K12 anyone can create, collaborate...

★ 1.0 • 4,453,892



Sketchboard

[sketchboardme](#)

Online whiteboard for real-time collaboration, and intuitive diagramming

★ 3.9 • 4,089,968



Colaboratory

This allows Google Colaboratory to open and create files in Google Drive. It is automatically installed on first use; uninstalling this will not prevent access to Colaboratory.

Install

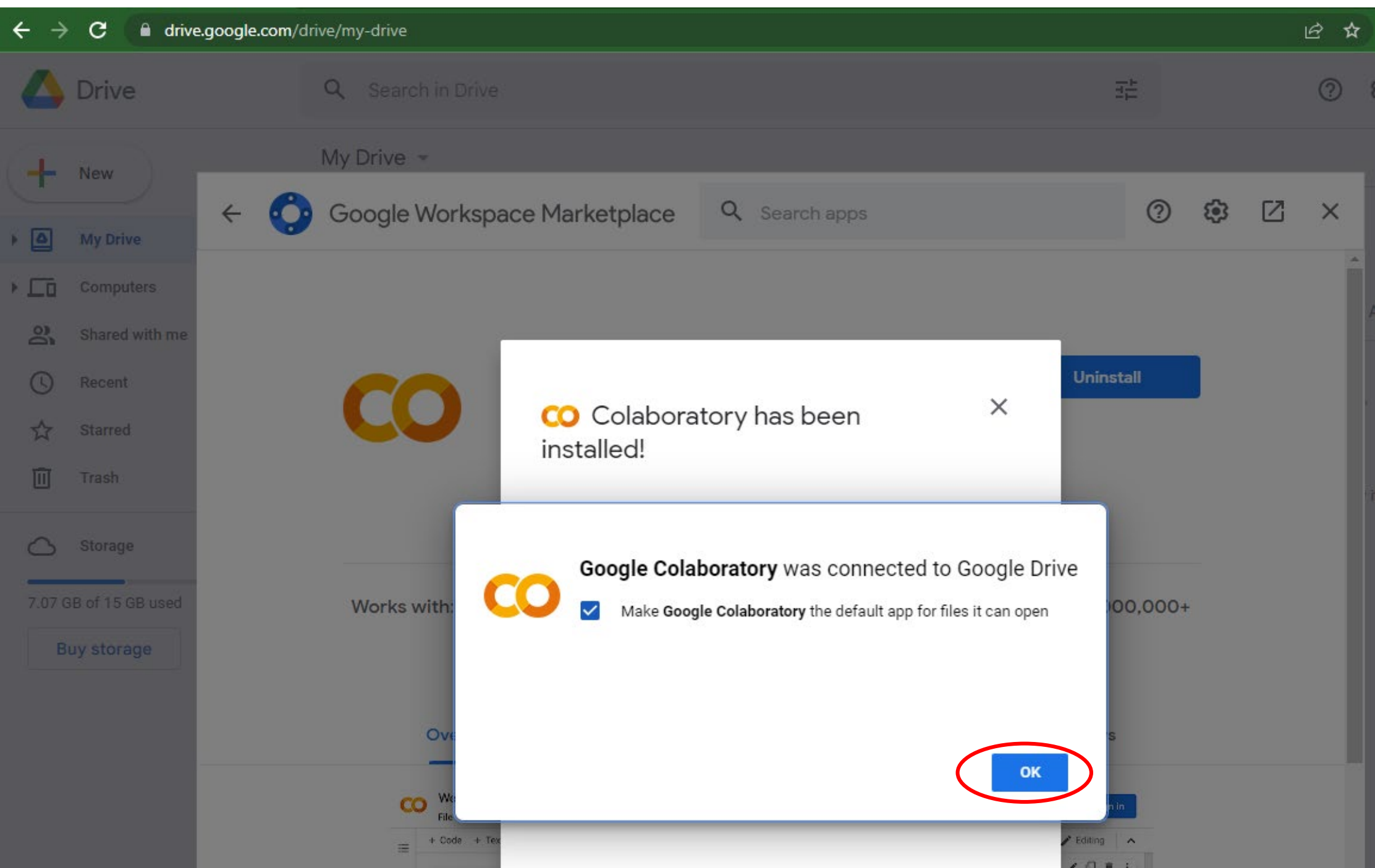
Get ready to install

Colaboratory needs your permission in order to start installing.

By clicking Continue, you acknowledge that your information will be used in accordance with the [terms of service](#) and [privacy policy](#) of this application.

CANCEL

CONTINUE



Choose
to
install
Colab



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Google Colaboratory

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My Drive - Google Drive



My Drive - Google Drive



Untitled0.ipynb - Colaboratory



colab.research.google.com/drive/1EHrKII9AZJZYOf6y3CLPmYJ_n-mJB_hw?authuser=3



Untitled0.ipynb



File

Edit

View

Insert

Runtime

Tools

Help

All changes saved

+ Code

+ Text



|





My Drive - Google Drive



My Drive - Google Drive



Untitled0.ipynb - Colabora



All About Us



colab.research.google.com/drive/1EHrKII9AZJZYOf6y3CLPmYJ_n-mJB_hw?authuser=3



Untitled0.ipynb



File Edit View Insert **Runtime** Tools Help All changes saved

+ Code + Text



```
if (5>2):  
    print("Five is greater than two!")
```



Five is greater than two!



+ Code

+ Text

Untitled0.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

Run all Ctrl+F9

Run before Ctrl+F8

Run the focused cell Ctrl+Enter

Run selection Ctrl+Shift+Enter

Run after Ctrl+F10

Interrupt execution Ctrl+M I

Restart runtime Ctrl+M .

Restart and run all

Factory reset runtime

Change runtime type

Manage sessions

View runtime logs

```
[1] if (5>2):  
    print("Five is  
    Five is greater t
```

You are beginning to le

```
✓ # This is a comm  
01 x=5  
y="John"  
# A variable nam  
# A variable nam  
# A variable nam  
# Variable names  
x = "awesome"  
print("Python is " + x)  
#Example  
x = 5  
y = 10  
print(x + y)  
print("Sum of x and y =",x+y)  
  
Python is awesome  
15  
Sum of x and y = 15
```


Notebook settings

Hardware accelerator

None



None

GPU

TPU



Output when saving this notebook

Cancel

Save

Notebook settings

Hardware accelerator

GPU 

To get the most out of Colab, avoid using a GPU unless you need one. [Learn more](#)

☐ Omit code cell output when saving this notebook

Cancel

Save


```
▶ if (5>2):  
    print("Five is greater than two!")
```

📄 Five is greater than two!

You are beginning to learn the Colab.

```
▶ # This is a comment.  
x=5  
y="John"  
# A variable name must start with a letter or the underscore character  
# A variable name cannot start with a number  
# A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _ )  
# Variable names are case-sensitive (age, Age and AGE are different variables)  
x = "awesome"  
print("Python is " + x)  
#Example  
x = 5  
y = 10  
print(x + y)  
print("Sum of x and y =",x+y)
```

```
Python is awesome  
15  
Sum of x and y = 15
```



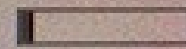
Comment



Share



RAM



Disk



Editing



Connect to a hosted runtime

Connect to a local runtime

View resources

Manage sessions

Show executed code history

Focus the Next Test Case



+ Code + Text



```
if (5>2):  
    print("Five is greater than two!")
```

Five is greater than two!

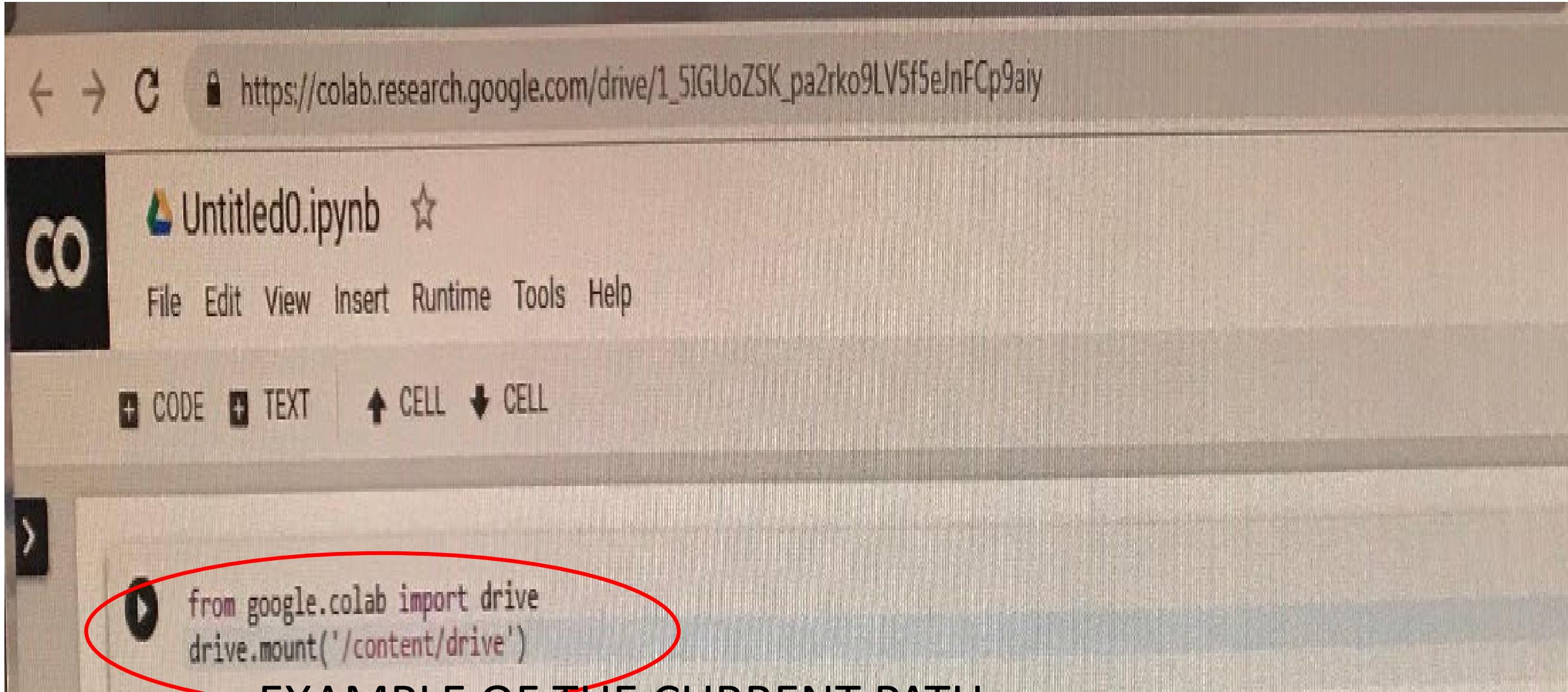
+ Code

+ Text

A new folder called
"Colab Notebooks" is created.

Python code runs section by section

Mount the Google Drive into the File space of Colab. Follow the link to confirm.



EXAMPLE OF THE CURRENT PATH:

```
data_raw = pd.read_csv('/content/drive/MyDrive/ELEC6604_Sept2022/Class4_folder/iris_dataset.csv', header = None, sep = ',')
```


+ Code + Text

File → Rename

```
[ ] if (5>2):  
    print("Five is greater than two!")  
  
Five is greater than two!
```

You are beginning to learn the Colab.

```
# This is a comment.  
x=5  
y="John"  
# A variable name must start with a letter or the underscore character  
# A variable name cannot start with a number  
# A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _ )  
# Variable names are case-sensitive (age, Age and AGE are different variables)  
x = "awesome"  
print("Python is " + x)  
#Example  
x = 5  
y = 10  
print(x + y)  
print("Sum of x and y =",x+y)
```

To run

```
Python is awesome  
15  
Sum of x and y = 15
```

Google Colaboratory is a free Jupyter notebook environment that runs on Google's cloud servers, with power of GPUs or TPUs. This lets you do everything you can in a Jupyter notebook hosted in your local machine, **without requiring the installations** and setup for hosting a notebook in your local machine.

Click to enter into the folder/directory “**Colab Notebooks**”.

Just **click-and-drag** to copy a file to the folder.

Limitations:

Colab is a temporary environment with an idle timeout of 90 minutes and an absolute timeout of 12 hours.

Colab has a disk space limitation of 108 GB, of which only 77 GB is available to the user.

Google Colab is a great tool for individuals who want to harness the power of high-end computing resources like GPUs, without being restricted by their price.